

Zabbix Monitoring System

Deployment & Troubleshooting Report

SecureBank SOC Lab

Infrastructure Monitoring Component

Component	Details
SIEM Server	192.168.44.10 (Ubuntu)
WEB Server	192.168.44.129 (Ubuntu 18.04)
Zabbix Version	6.0 LTS
Date	May 20, 2026

Comprehensive documentation of problems encountered and solutions applied during Zabbix server and agent deployment

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1. Introduction

This report documents the complete deployment process of the Zabbix monitoring system within the SecureBank SOC lab environment. The deployment involved installing and configuring Zabbix Server on the SIEM machine (192.168.44.10) and Zabbix Agent on the WEB server (192.168.44.129).

Lab Architecture

Component	IP Address	OS	Role
SIEM Server	192.168.44.10	Ubuntu 20.04+	Zabbix Server + Wazuh Manager
WEB Server	192.168.44.129	Ubuntu 18.04	Zabbix Agent + Wazuh Agent

Deployment Objectives

- Install and configure Zabbix Server 6.0 on SIEM
- Install and configure Zabbix Agent on WEB server
- Establish secure agent-server communication
- Configure host monitoring with Linux OS template
- Verify data collection and dashboard functionality

2. Problem 1: Zabbix Admin Account Locked

Problem Encountered

After initial Zabbix installation, attempting to log in with default credentials (Admin/zabbix) resulted in account lockout due to multiple failed attempts and incorrect database configuration.

Symptoms

- Login page displayed: *"Incorrect user name or password or account is temporarily blocked"*
- Multiple failed login attempts triggered security lockout
- Database connection issues prevented authentication

Figure 1: Zabbix Login Error (image(1).png)

Error: **"Incorrect user name or password or account is temporarily blocked"**

Screenshot shows the Zabbix login page with red error text below the username field.

Solution Applied

Step 1: Reset Admin Password and Unlock Account in MySQL

Connect to MySQL as root and execute the following SQL commands:

```
USE zabbix;
UPDATE users SET attempt_failed = 0,
               attempt_clock = 0,
               passwd = MD5('zabbix')
```

```
WHERE username = 'Admin';  
FLUSH PRIVILEGES;  
EXIT;
```

Step 2: Verify Frontend Configuration

Ensure the Zabbix frontend configuration file contains correct database credentials:

```
<?php  
$DB['TYPE']      = 'MYSQL';  
$DB['SERVER']    = 'localhost';  
$DB['PORT']      = '0';  
$DB['DATABASE']  = 'zabbix';  
$DB['USER']      = 'zabbix';  
$DB['PASSWORD']  = 'SecureBank123!';  
$DB['SCHEMA']    = '';  
$ZBX_SERVER      = 'localhost';  
$ZBX_SERVER_PORT = '10051';  
$ZBX_SERVER_NAME = 'SecureBank SOC';  
$IMAGE_FORMAT_DEFAULT = IMAGE_FORMAT_PNG;  
?>
```

File location: `/etc/zabbix/web/zabbix.conf.php`

Step 3: Restart Services

```
sudo systemctl restart zabbix-server zabbix-agent apache2
```

Important Note

The password `SecureBank123!` was chosen as the database password during initial MySQL setup. The frontend config must match this exactly.

3. Problem 2: Database Connection Error

Problem Encountered

After fixing the login block, a database error appeared: *"Access denied for user 'zabbix'@'localhost' (using password: YES)"*. This occurred because the frontend configuration file had an incorrect password.

Symptoms

- Database error page displayed after login attempt
- Error: *"Access denied for user 'zabbix'@'localhost' (using password: YES)"*
- Frontend config file had wrong password (zabbix123 instead of SecureBank123!)

Figure 2: Database Connection Error (image(2).png)

Error: "Access denied for user 'zabbix'@'localhost' (using password: YES)"

Screenshot shows a red-bordered error box with database access denied message.

Solution Applied

Root Cause

Two configuration files were created with conflicting passwords:

- `/etc/zabbix/zabbix.conf.php` — contained wrong password `zabbix123`
- `/etc/zabbix/web/zabbix.conf.php` — needed correct password `SecureBank123!`

Fix Commands

```
# Remove wrong config files
sudo rm -f /etc/zabbix/zabbix.conf.php
sudo rm -f /etc/zabbix/web/zabbix.conf.php

# Create correct config with proper password
sudo tee /etc/zabbix/web/zabbix.conf.php << 'EOF'
<?php
$DB['TYPE']      = 'MYSQL';
$DB['SERVER']    = 'localhost';
$DB['PORT']      = '0';
$DB['DATABASE']  = 'zabbix';
$DB['USER']      = 'zabbix';
$DB['PASSWORD']  = 'SecureBank123!';
$DB['SCHEMA']    = '';
$ZBX_SERVER     = 'localhost';
$ZBX_SERVER_PORT = '10051';
$ZBX_SERVER_NAME = 'SecureBank SOC';
$IMAGE_FORMAT_DEFAULT = IMAGE_FORMAT_PNG;
EOF

# Fix permissions
sudo chown www-data:www-data /etc/zabbix/web/zabbix.conf.php
sudo chmod 644 /etc/zabbix/web/zabbix.conf.php

# Create symlink for alternate location
sudo ln -sf /etc/zabbix/web/zabbix.conf.php /usr/share/zabbix/conf/zabbix.conf.php

# Restart Apache
sudo systemctl restart apache2
```

Verification

```
php -r 'echo mysqli_connect("localhost", "zabbix",
    "SecureBank123!", "zabbix") ? "DB OK" : "DB FAIL";'
# Output: DB OK
```

4. Problem 3: Zabbix Agent Installation on Ubuntu

18.04

Problem Encountered

The WEB server runs Ubuntu 18.04 (Bionic Beaver) with libc6 version 2.27. Zabbix Agent2 6.0 requires libc6 ≥ 2.28 (available only in Ubuntu 20.04+), causing dependency conflicts during installation.

Error Output

```
The following packages have unmet dependencies:
zabbix-agent2 : Depends: libc6 (>= 2.28)
                  but 2.27-3ubuntu1.5 is to be installed
E: Unable to correct problems, you have held broken packages.
```

Solution Applied

Solution: Use Zabbix Agent v1 (Compatible with Ubuntu 18.04)

Zabbix Agent v1 is fully compatible with Ubuntu 18.04 and provides all necessary monitoring capabilities.

```
# Install Zabbix Agent v1
sudo apt update
sudo apt install -y zabbix-agent

# Verify installation
dpkg -l | grep zabbix
ls -la /etc/zabbix/
```

Configuration

Edit `/etc/zabbix/zabbix_agentd.conf` :

```
# Set Zabbix Server IP
Server=192.168.44.10

# Set active server IP
ServerActive=192.168.44.10

# Set unique hostname (must match Zabbix dashboard)
Hostname=web01-securebank
```

Start and Enable Agent

```
sudo systemctl restart zabbix-agent
sudo systemctl enable zabbix-agent
sudo systemctl status zabbix-agent
```

Configuration Detail

Parameter	Value
Server	192.168.44.10
ServerActive	192.168.44.10
Hostname	web01-securebank
ListenPort	10050

5. Problem 4: Host Configuration and Template Assignment

Problem Encountered

After agent installation, the host appeared in Zabbix but showed gray availability (ZBX gray), indicating no active monitoring. The template "Template OS Linux by Zabbix agent" was not assigned, and a duplicate host with wrong IP existed.

Initial State

- Host "VM-WEB SecureBank" existed with wrong IP (192.168.1.199)
- Host "WEB Server - SecureBank" had correct IP (192.168.44.129) but gray status
- No monitoring template was assigned

Figure 3: Initial Host Status (image(3).png)

Host	IP	Availability
VM-WEB SecureBank	192.168.1.199:10050	RED
WEB Server - SecureBank	192.168.44.129:10050	GRAY
Zabbix server	127.0.0.1:10050	GREEN

Solution Applied

Step 1: Remove Incorrect Host

1. Go to Configuration → Hosts
2. Select "VM-WEB SecureBank" (checkbox)
3. Click Delete

Step 2: Configure Correct Host

Edit "WEB Server - SecureBank":

Figure 4: Host Configuration Form (image(4).png)

Screenshot shows the host edit form with fields for Host name, Visible name, Templates, Groups, and Interfaces.

Required Settings

Field	Value
Host name	web01-securebank
Visible name	WEB Server - SecureBank
Groups	Linux servers
Interface Type	Agent
IP Address	192.168.44.129
Port	10050
Template	Template OS Linux by Zabbix agent

Step 3: Verify Agent Connectivity

From SIEM server:

```
# Test agent response
zabbix_get -s 192.168.44.129 -k agent.ping
# Expected output: 1

zabbix_get -s 192.168.44.129 -k system.hostname
# Expected output: ubuntu
```

6. Verification and Final Status

Successful Monitoring

After applying all fixes, the host availability turned **GREEN**.

Figure 5: Final Host Status (image(5).png)

Host	IP	Availability
WEB Server - SecureBank	192.168.44.129:10050	ZBX GREEN
Zabbix server	127.0.0.1:10050	ZBX GREEN

Dashboard Overview

Figure 6: Zabbix Global Dashboard (image(6).png)

Screenshot shows the Zabbix Global View dashboard with System Information, Available/Not Available/Unknown host counts, Problems list, and Favorite graphs section.

Widget Configuration

Figure 7: Add Widget Dialog (image(7).png)

Screenshot shows the "Add widget" dialog with Type set to "Graph", Name field showing "WEB Server - SecureBank", and Data set configuration options.

7. File Integrity Monitoring with Wazuh

Problem Encountered

Attempting to configure Wazuh File Integrity Monitoring (FIM) for `/var/www/html` encountered multiple configuration syntax errors, causing the Wazuh agent to fail startup with "Invalid element in the configuration: 'syscheck'".

Error Sequence

1. Duplicate `<syscheck>` sections created
2. XML syntax corrupted with `<<` instead of `<`
3. All syscheck sections accidentally deleted
4. Agent failed with: "No directory provided for syscheck to monitor"

Figure 8: Wazuh FIM Alerts in SIEM (image(8).png)

Terminal screenshot showing Wazuh alerts JSON output with sudo events and FIM rule IDs (550, 553, 554) for file add/modify/delete detection.

Figure 9: Wazuh Agent Status - Broken Config (image(9).png)

Agent running but FIM disabled: "No directory provided for syscheck to monitor"

Systemctl status output showing wazuh-syscheckd INFO messages indicating FIM is disabled due to missing directory configuration.

Figure 10: Reinstall Attempt with Same Issue (image(10).png)

Reinstalled agent still shows: "File integrity monitoring disabled"

Terminal showing wazuh-agent reinstall and restart, with same syscheck error persisting due to sed command corruption.

Solution Applied

Root Cause

The `sed` commands used double angle brackets (`<<`) instead of single brackets (`<`) for XML tags, corrupting the configuration file.

Correct Configuration

The `<syscheck>` section must contain:

```
<syscheck>
  <disabled>no</disabled>
  <frequency>300</frequency>
  <scan_on_start>yes</scan_on_start>
  <directories>/etc,/usr/bin,/usr/sbin</directories>
  <directories>/bin,/sbin,/boot</directories>
  <directories>/var/www/html</directories>
  <ignore>/etc/mtab</ignore>
  <ignore>/etc/hosts.deny</ignore>
  <ignore>/etc/mail/statistics</ignore>
</syscheck>
```

Recommended Fix Approach

```
# Backup current config
sudo cp /var/ossec/etc/ossec.conf /var/ossec/etc/ossec.conf.backup

# Use tee with heredoc to avoid sed escaping issues
sudo bash -c 'cat > /var/ossec/etc/ossec.conf << EOF
<ossec_config>
  <client>
    <server>
      <address>192.168.44.10</address>
      <port>1514</port>
      <protocol>udp</protocol>
    </server>
    <config-profile>ubuntu, ubuntu18, debian</config-profile>
    <auto_restart>yes</auto_restart>
  </client>

  <syscheck>
    <disabled>no</disabled>
    <frequency>300</frequency>
    <scan_on_start>yes</scan_on_start>
    <directories>/etc,/usr/bin,/usr/sbin</directories>
    <directories>/bin,/sbin,/boot</directories>
    <directories>/var/www/html</directories>
    <ignore>/etc/mtab</ignore>
    <ignore>/etc/hosts.deny</ignore>
  </syscheck>

  <localfile>
    <log_format>syslog</log_format>
    <location>/var/log/auth.log</location>
  </localfile>
</ossec_config>
EOF'
```

```
# Restart agent
sudo systemctl restart wazuh-agent
sudo systemctl status wazuh-agent
```

Important Note

FIM Rule IDs in Wazuh:

- **Rule 550:** File added to system
- **Rule 553:** File modified
- **Rule 554:** File deleted

These rules generate alerts visible in Kibana under `rule.groups: syscheck`.

8. Summary of All Solutions

Complete Problem-Solution Matrix

Problem	Root Cause	Solution
Admin account locked	Too many failed login attempts	Reset password in MySQL: <code>UPDATE users SET attempt_failed=0, passwd=MD5('zabbix')</code>
Database access denied	Wrong password in frontend config	Recreate <code>zabbix.conf.php</code> with correct password SecureBank123!
Agent2 install fails	Ubuntu 18.04 has <code>libc6 2.27</code> , needs 2.28	Use Zabbix Agent v1: <code>apt install zabbix-agent</code>
Host shows gray status	No template assigned	Add "Template OS Linux by Zabbix agent"
Wrong host exists	Duplicate with incorrect IP	Delete "VM-WEB SecureBank" (IP 192.168.1.199)
Wazuh FIM disabled	Corrupted XML config from <code>sed</code>	Use <code>tee</code> with <code>heredoc</code> , avoid <code>sed</code> for XML

Final System Status

Component	Host	Status	IP
Zabbix Server	SIEM	Running	192.168.44.10
Zabbix Agent v1	WEB	Running	192.168.44.129
Wazuh Manager	SIEM	Running	192.168.44.10

Wazuh Agent	WEB	Running	192.168.44.129
MySQL Database	SIEM	Connected	localhost
Apache Web Server	SIEM	Running	localhost

Access URLs

- **Zabbix Dashboard:** <http://192.168.44.10/zabbix>
- **Kibana (Wazuh):** <http://192.168.44.10:5601>
- **Zabbix Login:** Admin / zabbix

Deployment Complete

SecureBank SOC Monitoring Infrastructure

Fully Operational as of May 20, 2026